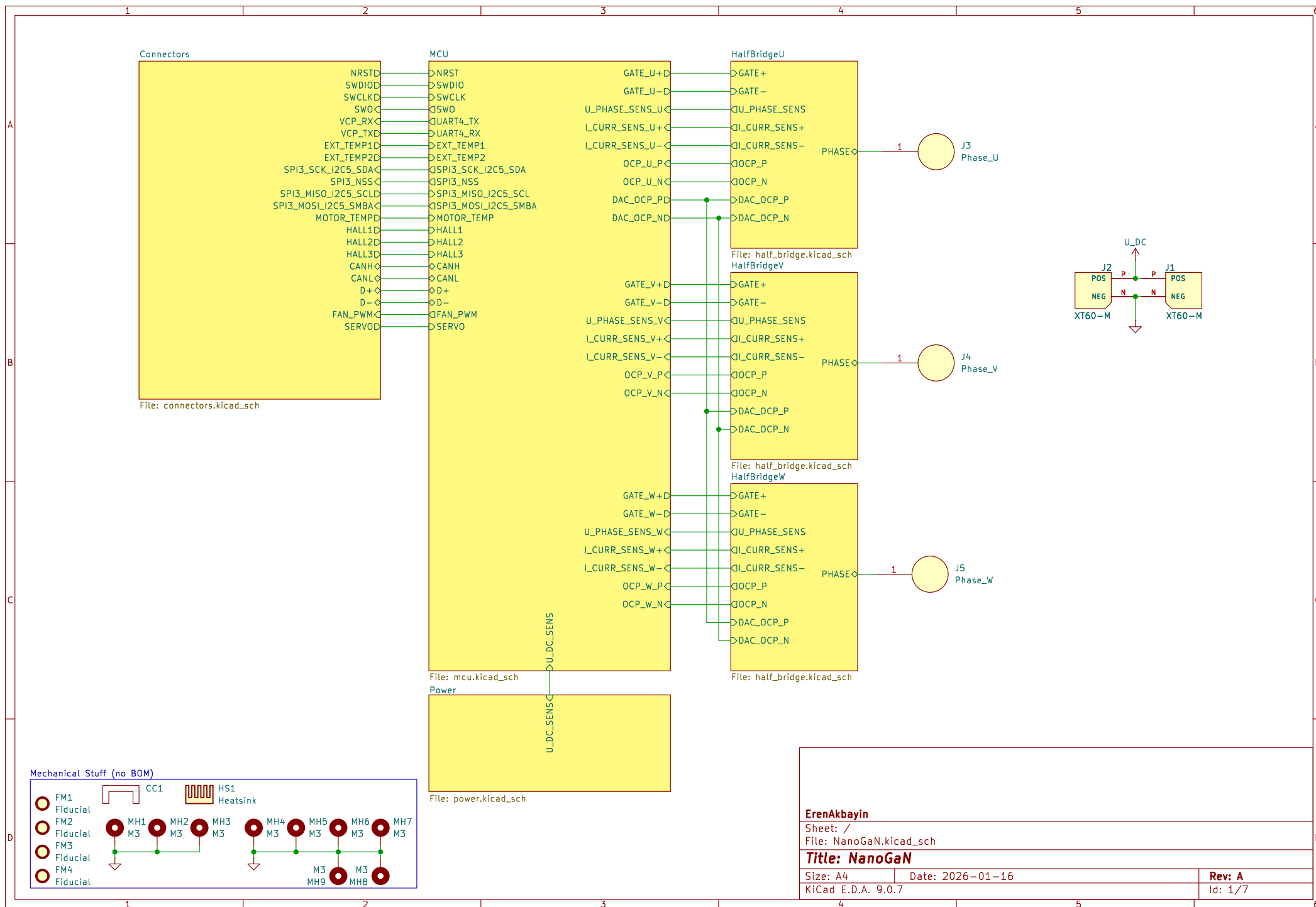
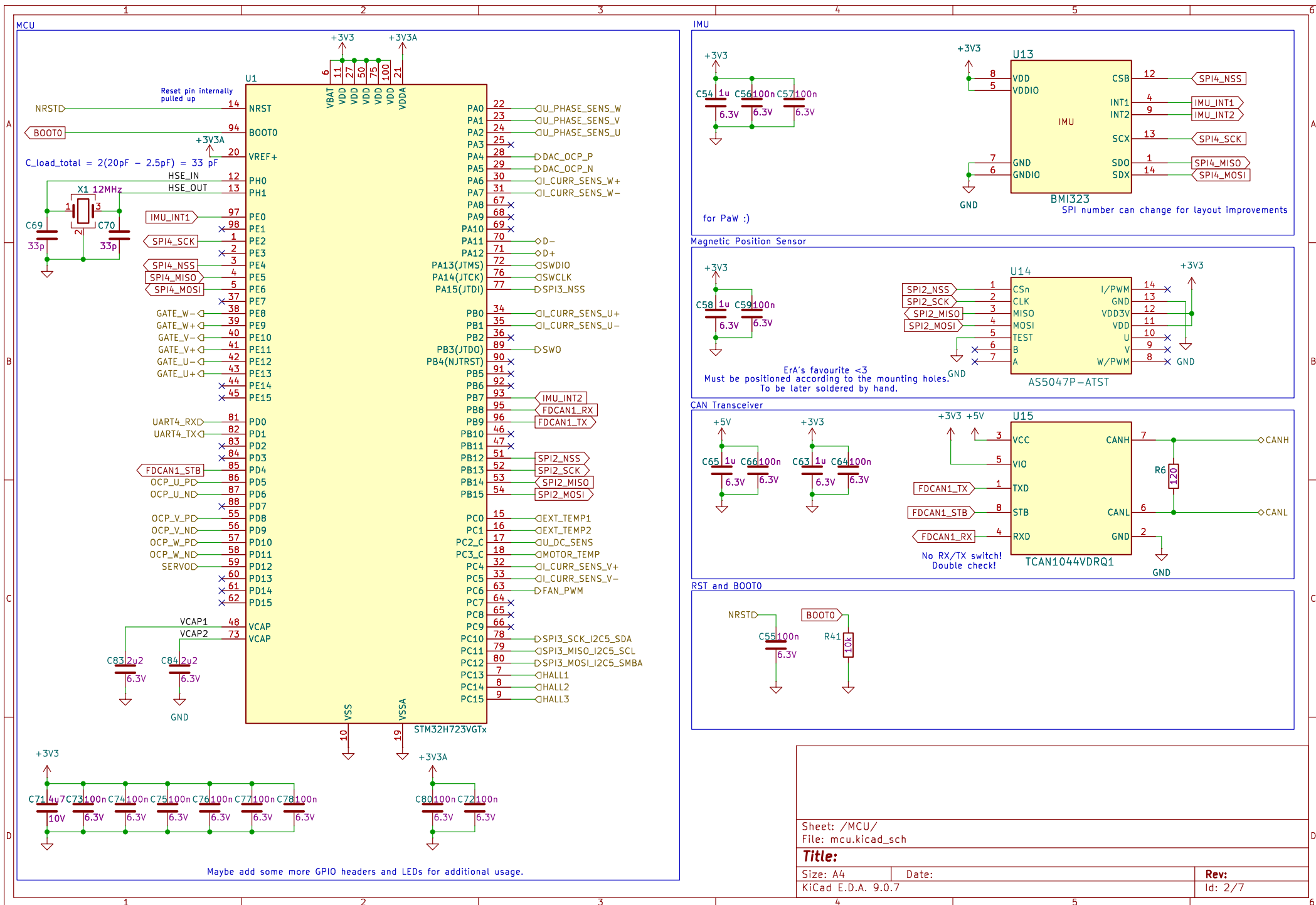
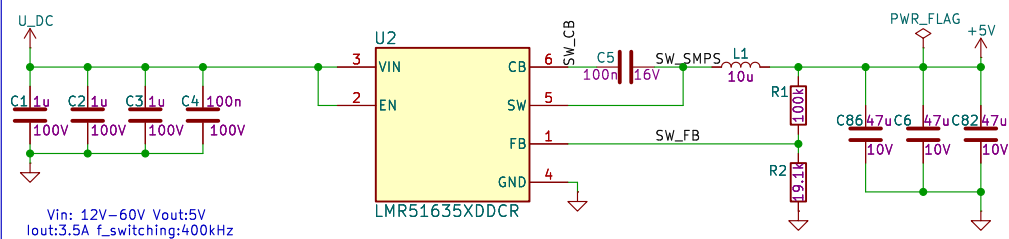






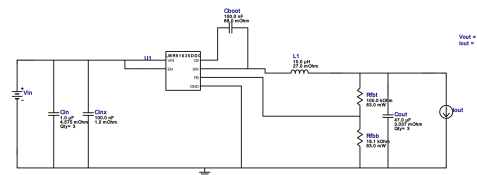
Buck Converter



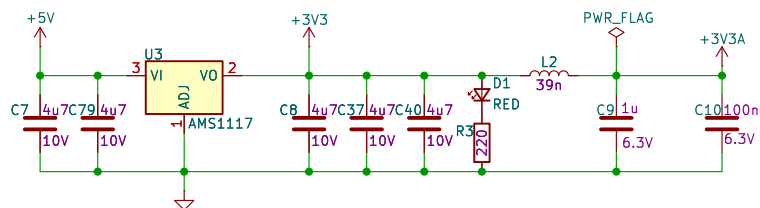
WEBENCH® Design Report

Design : 29 LMR51635DDC
LMR51635DDC 12V-60V to 5.00V @ 3.5A

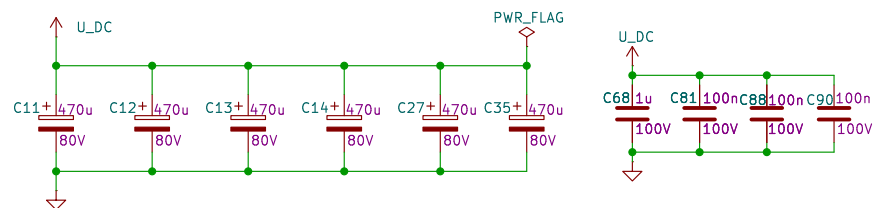
VinMin = 12.0V
VinMax = 60.0V
Vout = 5.0V
Iout = 3.5A
Device = LMR51635DDC
Topology = Buck
Created = 2026-01-18 17:57:50.631
BOM Cost = \$2.43
BOM Count = 12
Total Pd = 2.47W



Low drop-out regulator & GNDA / GND separation

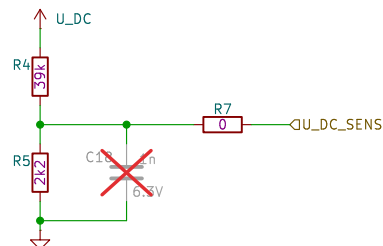


Bulk Capacitors



Extra ceramic (X7R) capacitors for improving switching performance.

DC-Link Voltage



Sheet: /Power/
File: power.kicad_sch

Title:

Size: A4

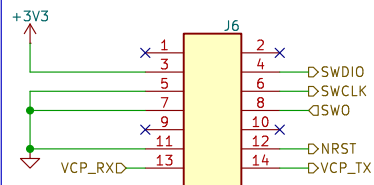
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KiCad E.D.A. 9.0.7

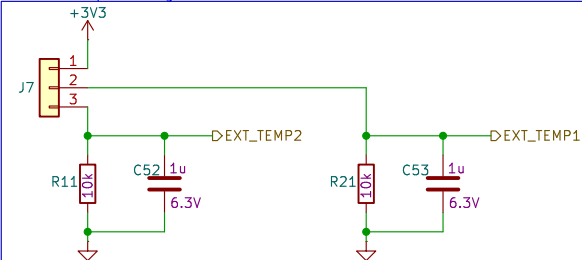
Rev:

Id: 3/7

ST-Link V3 MINIE CONNECTOR (STDC14)

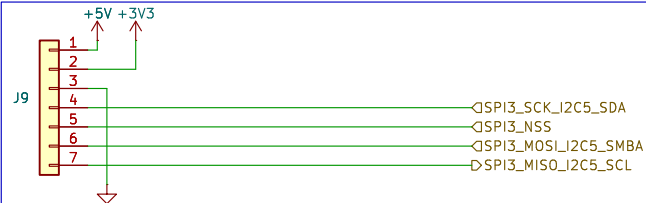


External Temp. Readings

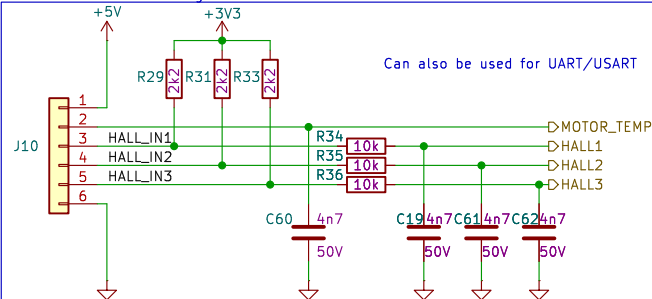


for motor and etc.

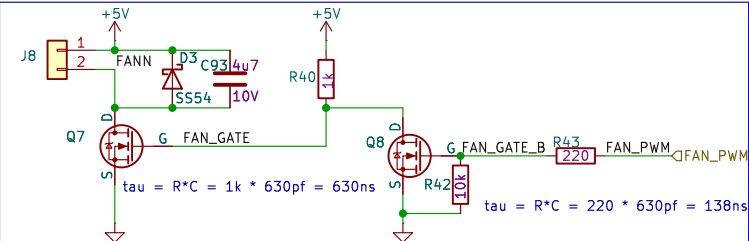
SPI/I2C/USART connection



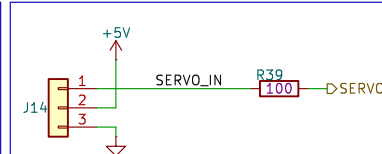
Hall connector and filtering



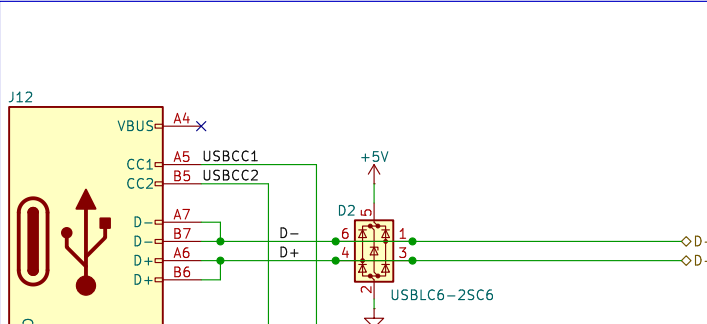
Fan connector



Servo



USB-C Connector



Differential Pair settings

Layers 4 PCB Thickness 0.8 Inner Copper Weight 0.3oz Outer Copper Weight 1oz Unit mm

Impedance Configure

+ New Impedance

+ Duplicate Impedance

+ Calculate

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Spacing (mm)	Impedance trace to copper (mm)
90	Differential Pair (Non-coupled)	L1	L2	L3	0.20	0.20
100	Differential Pair (Non-coupled)	L1	L2	L3	0.20	0.20

JLC0808H-3313 (Standard) (Finished Thickness: 0.745-0.8mm)

JLC0808H-1080 (Special) (Finished Thickness: 0.59-0.6mm)

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width (mm)	Trace Spacing (mm)	Impedance trace to copper (mm)
90	Differential Pair (Non-coupled)	L1	L2	L3	0.20	0.20	0.20
100	Differential Pair (Non-coupled)	L1	L2	L3	0.20	0.20	0.20

Layer	Material	Thickness (mm)	Thickness (mm)
L1	Outer Copper Weight	1.38	0.0200
Prepreg	3313 RC27% 4.2mil	3.91	0.0994
L2	Inner Copper Weight	0.60	0.0152
Core	0.075mm VMCZ without copper	2.95	0.0750
L3	Inner Copper Weight	0.60	0.0152
Prepreg	3313 RC27% 4.2mil	3.91	0.0994
L4	Inner Copper Weight	0.60	0.0152
Core	0.075mm VMCZ without copper	2.95	0.0750
L5	Inner Copper Weight	0.60	0.0152
Prepreg	3313 RC27% 4.2mil	3.91	0.0994
L6	Outer Copper Weight	1.38	0.0200

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File: connectors.kicad_sch

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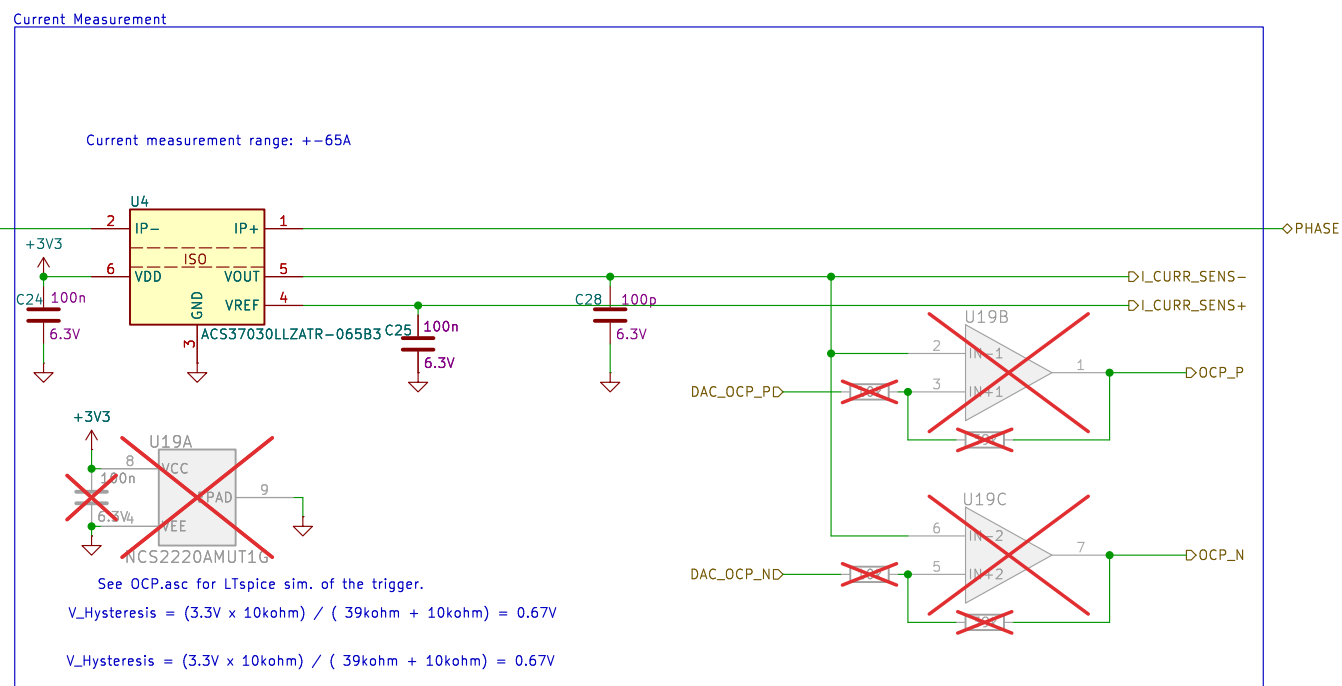
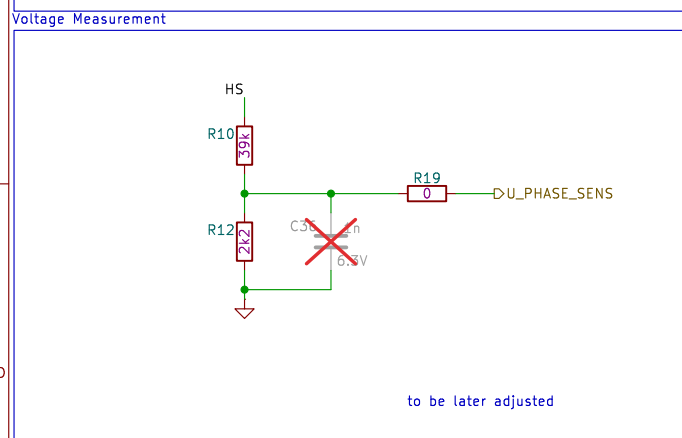
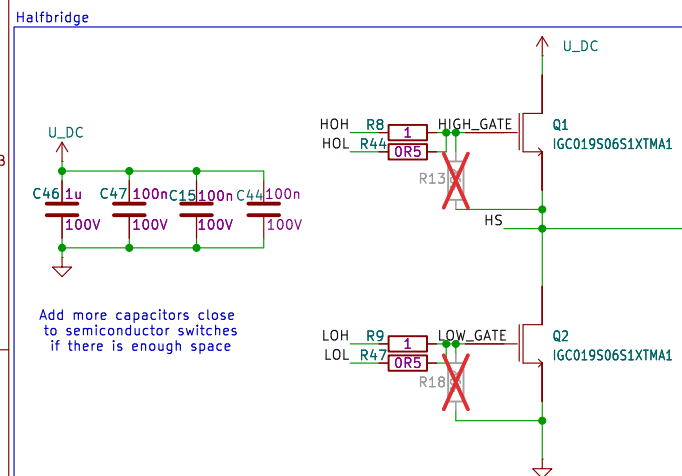
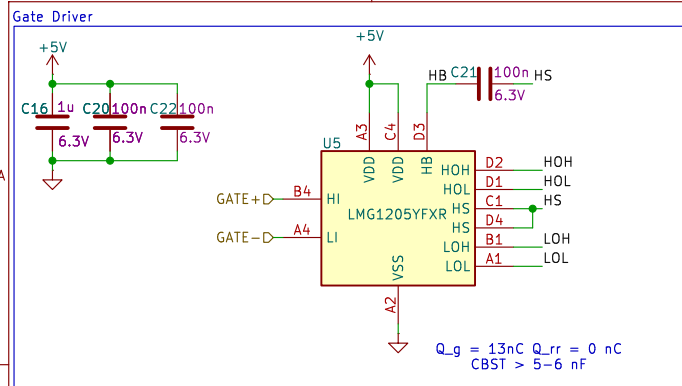
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Date:

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Rev:

Id: 4/7



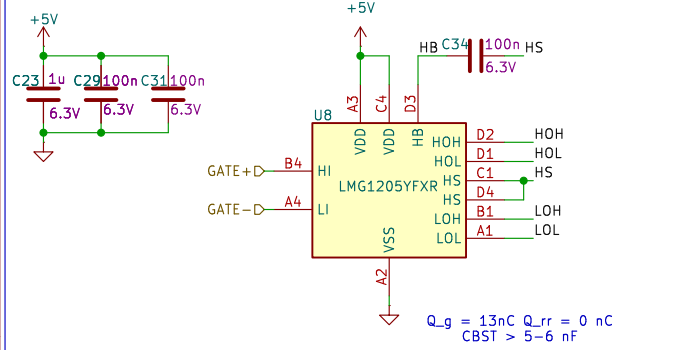
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File: half_bridge.kicad_sch

Title:

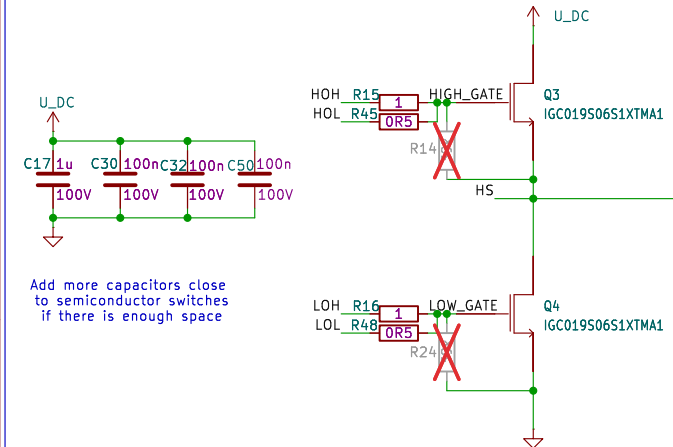
Size: A4	Date:
KiCad E.D.A. 9.0.7	

Rev:
Id: 5/7

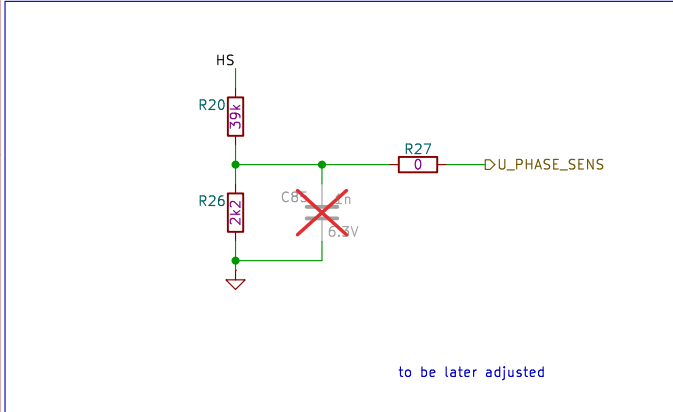
Gate Driver



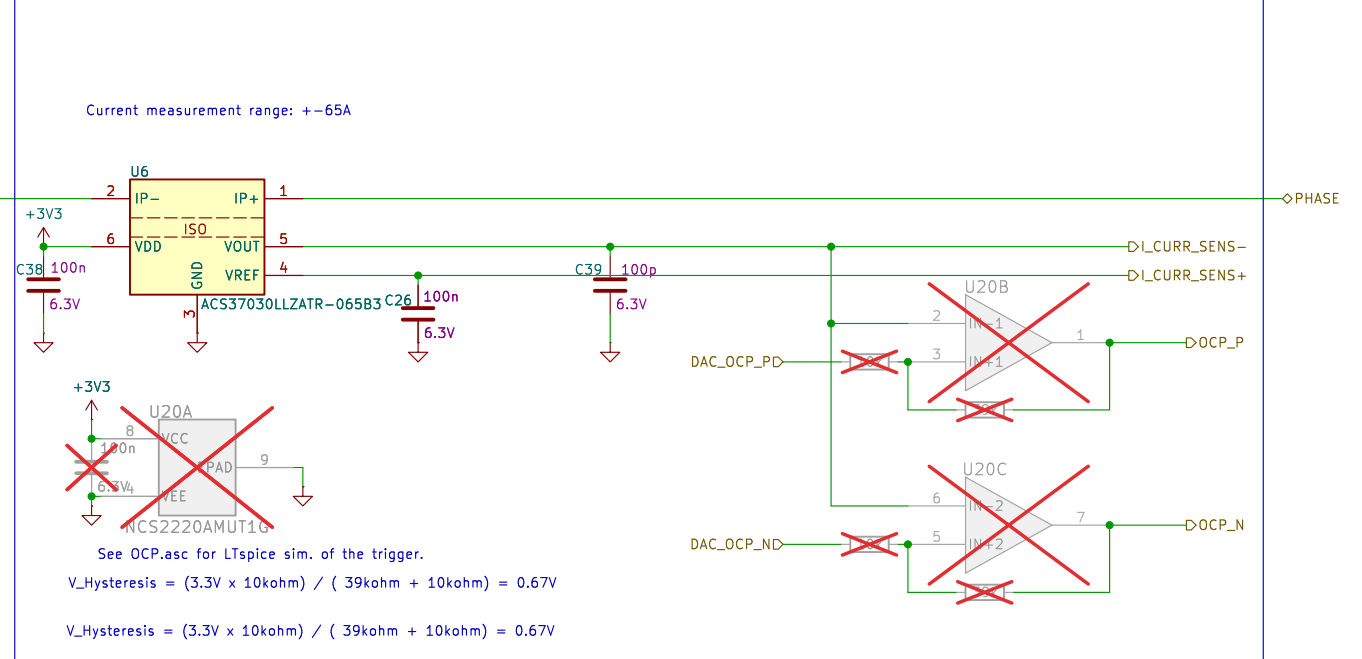
Halfbridge



Voltage Measurement



Current Measurement



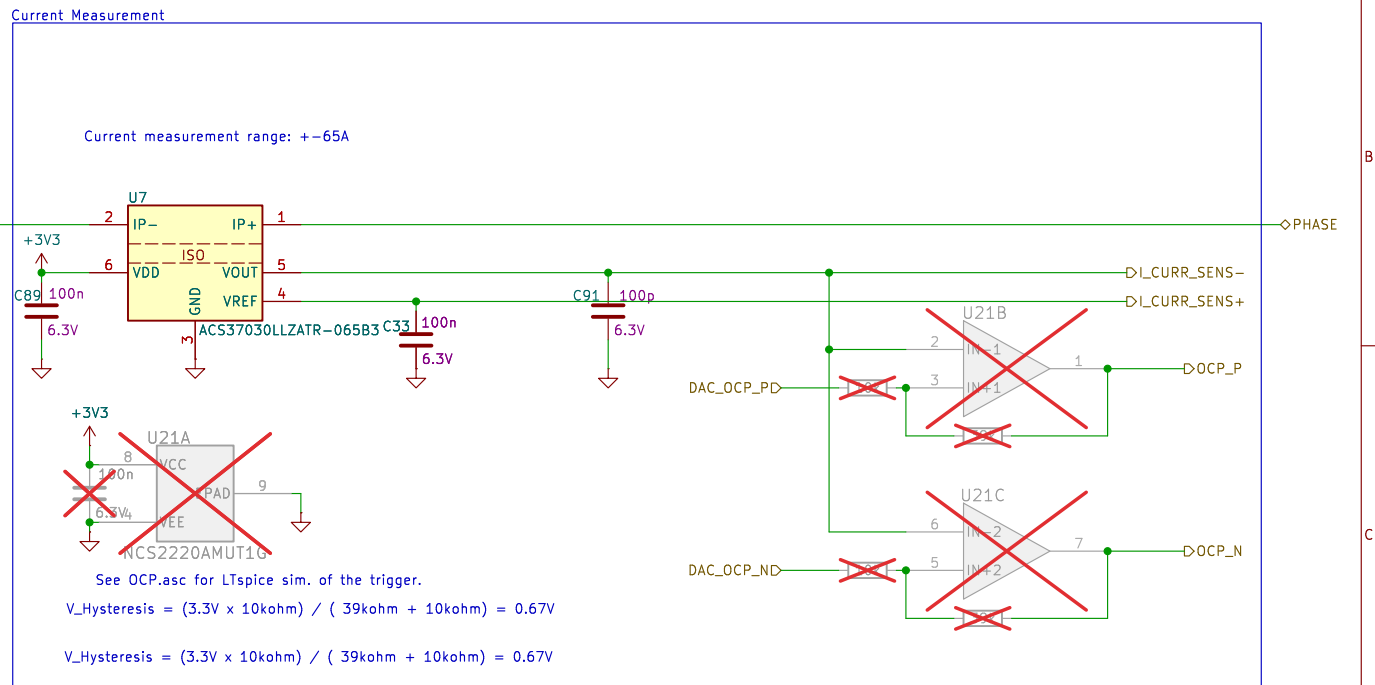
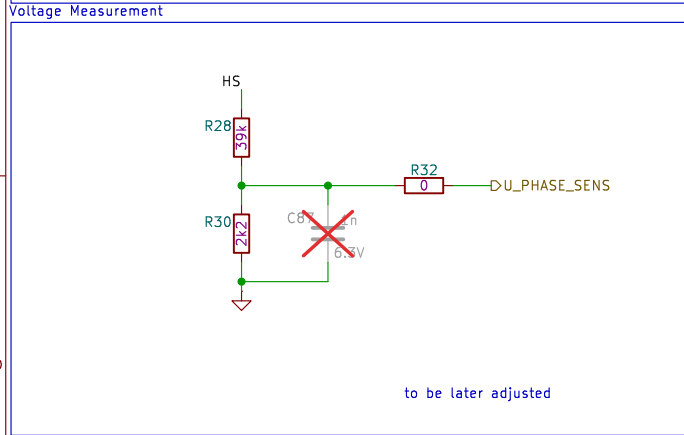
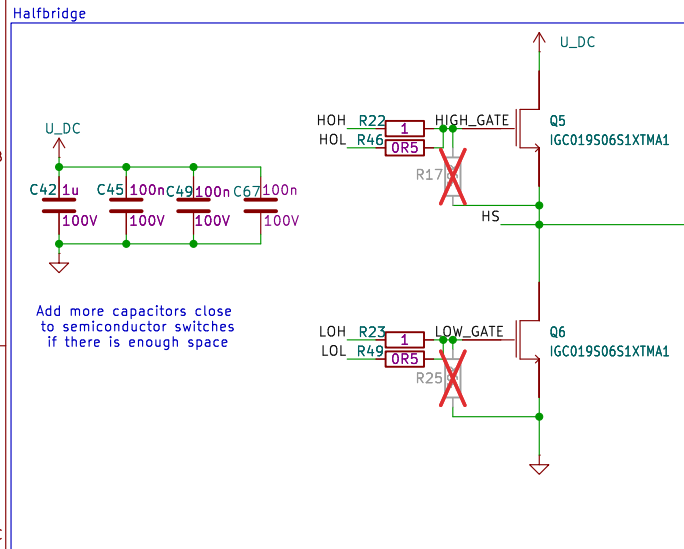
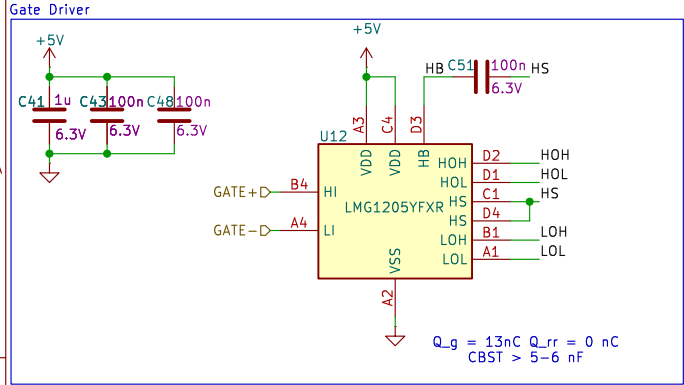
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Title:

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev:
Id: 6/7



Sheet: /HalfBridgeW/
File: half_bridge.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.7

Date:

Rev:
Id: 7/7